NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER 1.

SATURDAY 17TH MARCH, 2018 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) *Answer scripts of only qualified students should be returned to the contest coordinator, latest* **Monday 19th March, 2018**.

(d) **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 28*th July,* 2018 *at NOON at Makerere University, Kampala.*

(e) *National Mathematics Contest* 2018 *Paper* 2 *will be done on* 23*rd June,* 2018 *at various centres.*

SECTION A : 5 marks each

1. There are three consecutive positive integers such that the square of the second minus twelve times the first is three less than twice the third. What is the smallest of the three integers?

2. In what positive base x does the equation $4 \times 12 = 103$ for multiplication of base-x numbers hold?

3. What positive integers less than 100 have exactly three distinct factors?

4. (a) For what positive integer n is $100^2 - n^2 = 9559$?
   (b) Determine one pair (x, y) of positive integers for which $x > 1$ and $y > 1$ and $xy = 9559$.

5. Let
$$s_n = 1 - 2 + 3 - 4 + 5 - 6 + \cdots + (-1)^{n+1} n.$$
Find $s_1 + s_2 + s_3 + \ldots + s_{2018}$.

6. What is the value of
$$(\log_3(\log_2(\log_4(65536))))^{2018}?$$

7. In triangle ABC, D lies on BC so that $AC = 4\ cm, AD = 4\ cm, BD = 8\ cm$, and $CD = 1\ cm$. What is the value of angle ABC?

8. A student has four shirts and three trousers. The four shirts are Red, Blue, Green and Yellow, and the three trousers are Red, Blue and Green. If the student picks a shirt and a trouser at random. What is the probability that they are of the same color?

9. In the Zagland, people measure length in two units, zigs and zags. A length of $2x$ zigs is equal to a length of $(2x+6)$ zags and a length of $(3x+7)$ zigs is equal to a length of $(3x+22)$ zags. what is the value of x?

10. The ratio of boys to girls in a class is $7:5$. If there are 6 more boys than girls, how many girls are in the class?

SECTION B : 10 marks each

11. If all people eat the same amount of pizza, and a pizza 16 inches in diameter serves two people, how many inches in diameter should each of two pizzas be in order to serve three people? (Pizzas are circular and are eaten entirely.)

12. How many pairs (x, y) satisfying $10 \leq y < x$ and $x + y \leq 99$ have the property that $x + y$ and $x - y$ have the same digits in reverse order? (This allows cases such as $(44; 36)$ where the sum and difference are 80 and 08.)

13. John and David are running laps around a 500-meter oval track. They started together, but John has pulled ahead because he is 20% faster than David. How many laps will John have run when he

    (a) first passes David?

    (b) passes David for the second time.

14. A certain sequence of numbers is defined by $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 4$ and n is a natural number. Suppose $a_4 = 20, a_5 = 36$, and $a_9 = 408$. What is the value of a_1?

15. Circle B passes through the center of circle A and is tangent to it. Circle C passes through the center of circle B and is tangent to it. What fraction of the area of circle A lies inside circle B but outside circle C?

SECTION A : 5 marks each

Qn. 1. There are three consecutive positive integers such that the square of the second minus twelve times the first is three less than twice the third. What is the smallest of the three integers?

solution

Let the numbers be $x, x+1$ and $x+2$. Then $(x+1)^2 - 12x = 2(x+2) - 3 \Rightarrow x(x-12) = 0$. So the smallest of the three numbers is either 0 or 12.

Qn. 2. In what positive base x does the equation $4 \times 12 = 103$ for multiplication of base-x numbers hold?

solution

Convert every expression to base ten. $4_x \times 12_x = 103_x$ is equivalent to

$$(4 \times x^0) \times [(1 \times x^1) + (2 \times x^0)] = (1 \times x^2) + (0 \times x^1)(3 \times x^0)$$

whence $(x-5)(x+1) = 0 \Rightarrow x = 0$ or $x = 5$ and the positive base therefore is 5.

Qn. 3. What positive integers less than 100 have exactly three distinct factors?

solution

A number to have three distinct factors has to be a perfect square and they are $4, 9, 25$ and 49 with the three distinct factors as $\{1,2,4\}, \{1,3,9\}, \{1,5,25\}$ and $\{1,7,49\}$ respectively.

Qn. 4. (a) For what positive integer n is $100^2 - n^2 = 9559$?

(b) Determine one pair (x, y) of positive integers for which $x > 1$ and $y > 1$ and $xy = 9559$.

solution

(a) $100^2 - n^2 = 9559 \Rightarrow n^2 = 100^2 - 9559 = 441$ and therefore $n = 21$.

(b) Using the fact in (a) above we have $100^2 - 21^2 = 9559$. Using a difference of two squares we get $100^2 - 21^2 = (100-21)(100+21)$ thus the numbers are $x = 121, y = 79$ and $x = 869, y = 11$ and the order could be interchanged.

Qn. 5. Let $s_n = 1 - 2 + 3 - 4 + 5 - 6 + \cdots + (-1)^{n+1}n$. Find $s_1 + s_2 + s_3 + \ldots + s_{2018}$.

solution

The sums $s_1 = 1, s_2 = -1, s_3 = 2, s_4 = -2, s_5 = 3, s_6 = -3, \ldots$ Their sums therefore are $s_1 + s_2 = 0, s_1 + s_2 + s_3 = 2, s_1 + s_2 + s_3 + s_4 = 0, \ldots$ This sum will be zero always for n even thus $s_1 + s_2 + \ldots + s_{2018} = 0$.

Qn. 6. What is the value of

$$(\log_3(\log_2(\log_4(65536))))^{2018}?$$

solution

The value of $(\log_3(\log_2(\log_4(65536))))^{2018}$ is obtained after evaluating $\log_4(65536)$. Since $4^8 = 65536$ then $\log_4(65536) = \log_4(4^8) = 8$. So $\log_2(8) = 3$ and hence $(\log_3(3))^{2018} = 1$.

Qn. 7. In triangle ABC, D lies on BC so that $AC = 4\ cm, AD = 4\ cm, BD = 8\ cm$, and $CD = 1\ cm$. What is the value of angle ABC?

solution

From the diagram, $4^2 = h^2 + (DE)^2 \Rightarrow \frac{63}{4}$. Now $(AB)^2 = h^2 + (BE)^2$ But $BE = 8 + 0.5 = 8.5cm$

So $AB = \sqrt{\left(\frac{63}{4} + \frac{17^2}{4}\right)} = \frac{\sqrt{352}}{2}cm$. From $\sin\theta = \frac{\sqrt{63}}{2} \div \frac{352}{2} = 0.4231$ so

$\theta = \sin^{-1}(0.4231) = 25.03^\circ$

Qn. 8. A student has four shirts and three trousers. The four shirts are Red, Blue, Green and Yellow, and the three trousers are Red, Blue and Green. If the student picks a shirt and a trouser at random. What is the probability that they are of the same color?

solution

Each of the four shirts has three possible trousers to wear with thus $4 \times 3 = 12$ possible matches irrespective of color. Matching colors has only three possibilities (Red shirt, Red trouser, Green shirt Green trouser and Blue shirt Blue trouser)thus the probability is $\frac{3}{12} = \frac{1}{4}$.

Qn. 9. In the Zagland, people measure length in two units, zigs and zags. A length of $2x$ zigs is equal to a length of $(2x + 6)$ zags and a length of $(3x + 7)$ zigs is equal to a length of $(3x + 22)$ zags. what is the value of x?

solution

$2x$ zigs= $(2x + 6)$ zags $\Rightarrow$ 1 zig $= \frac{x+3}{x}$ zags. Now $(3x + 7)$ zigs $= (3x + 7)\frac{x+3}{x} = 3x + 22$ zags therefore $x = \frac{7}{2}$.

Qn. 10. The ratio of boys to girls in a class is $7 : 5$. If there are 6 more boys than girls, how many girls are in the class?

solution

Let the number of girls in the class be g. Then the number of boys in the class is $g + 6$ so $\frac{g+6}{g+(g+6)} = \frac{7}{7+5} \Rightarrow g = 15$.

SECTION B : 10 **marks each**

Qn. 11. If all people eat the same amount of pizza, and a pizza 16 inches in diameter serves two people, how many inches in diameter should each of two pizzas be in order to serve three people? (Pizzas are circular and are eaten entirely.)

solution

Let the diameter of each pizza be d inches. Then the total amount of pizza in the 2 pizzas is

$\pi \times \left(\frac{d}{2}\right)^2 \times 2 = \frac{\pi d^2}{2}$. Pizza eaten by one person $= \frac{\pi 8^2}{2}$ so pizza eaten by 3 persons is $3 \times \frac{\pi 8^2}{2}$

therefore $3 \times \frac{\pi 8^2}{2} = \frac{\pi d^2}{2} \Rightarrow d = 8\sqrt{3}$.

Qn. 12. How many pairs (x, y) satisfying $10 \leq y < x$ and $x + y \leq 99$ have the property that $x + y$ and $x - y$ have the same digits in reverse order? (This allows cases such as $(44; 36)$ where the sum and difference are 80 and 08.)

solution

Let $x + y = 10n + m$ and $x - y = 10m + n$ where $10m + n$ the largest value that the sum and difference can take on. Adding these equations gives $2x = 11(n + m)$ meaning x is a multiple of 11 and possible values are $x = \{11, 22, 33, 44, 55, 66, 77, 88, 99\}$ subtracting the equations yields $2y = 9(n - m)$ thus y is a multiple of 9 whose possible values are $y = \{9, 18, 27, 36, 45, 54, 63, 70, 81, 90, 99\}$. With restrictions $10 \leq y < x$ and $x + y \leq 99$ then there are **twelve pairs** satisfying thus
$\{(x, y) = (22, 18), (33, 18), (33, 27), (44, 18), (44, 27), (44, 36), (55, 18), (55, 27), (55, 36), (66, 18), (66, 27), (77, 18)\}$

Qn. 13. John and David are running laps around a 500-meter oval track. They started together, but John has pulled ahead because he is 20% faster than David. How many laps will John have run when he

(a) first passes David?

(b) passes David for the second time.

solution

Let the number of laps John will have run when he first passes David be l and the number of laps David will have run then be m. Then John will have covered $500l$ meters and David will have covered $500m$ meters. Suppose John is running at vms^{-1} then John is running at $1.2v\ ms^{-1}$. Since at the point John passes David they will both have used the sametime then $\frac{500l}{1.2v} = \frac{500m}{v} \rightarrow l = 1.2m$ that is $\frac{6}{5}$. Since full laps have to be whole numbers,

(a) David will have done 5 laps and John 6 by the time John first passes David.

(b) John will pass David the second time when John has done 12 laps and David 10.

Qn. 14. A certain sequence of numbers is defined by $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 4$ and n is a natural number. Suppose $a_4 = 20, a_5 = 36$, and $a_9 = 408$. What is the value of a_1?

solution

Write down equations out of the sequence for $n = 4$ and $n = 5$ that is $a_1 + a_2 + a_3 = a_4 = 20$ and $a_2 + a_3 + a_4 = 36$. solving the two yields $a_1 = 4$.

Qn. 15. Circle B passes through the center of circle A and is tangent to it. Circle C passes through the center of circle B and is tangent to it. What fraction of the area of circle A lies inside circle B but outside circle C?

solution

Fraction of the area required $= \frac{\text{shaded area}}{\text{area of } A}$. Let r be the radius of the circle C then radius of circle B is $2r$ and radius A is $4r$. Fraction of the area required $= \frac{\pi(2r)^2 - \Pi r^2}{\pi(4r)^2} = \frac{3}{16}$.